

Introduction to I2C Protocol

Ankit Srivastava
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OVERVIEW

- Characteristics and features of I2C
- How I2C Protocol works?

I2C CHARACTERISTICS

- I2C (also written as I²C) stands for Inter-Integrated Circuit
- I2C characteristics:
 - Uses just two wires + ground
 - Bus-based
 - Multi-master
 - Low-bandwidth
 - Synchronous
 - Short-distance
 - Serial
 - Uses in-band addressing
 - Half-duplex

I2C CHARACTERISTICS

- Only a master device can initiate and terminate the data transfer
- True multi-master bus including collision detection & arbitration to prevent data corruption if two or more masters simultaneously initiate data transfer
 - We only consider the single master case here, i.e., STM32 chip is always the only master. Hence, we will not discuss arbitration
- Master generates the clock irrespective of the stage & direction of transfer
- During transfer, any device addressed by the master is considered a slave, recognized by a unique 7-bit address (a 10-bit variant exists)
- The number of devices that can be connected to the same bus is limited only by the maximum bus capacitance

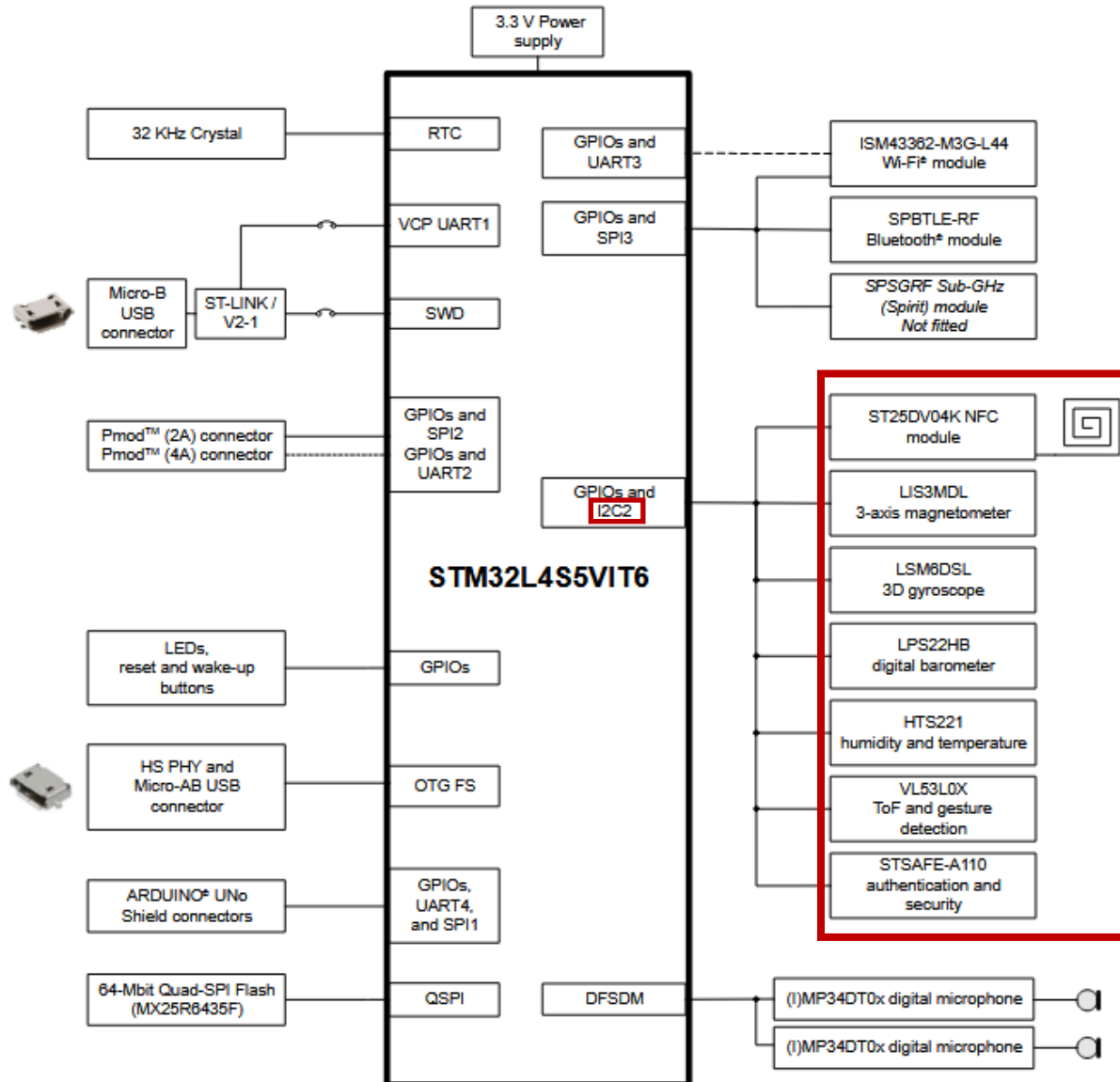
I2C CHARACTERISTICS

- Originally developed by Philips, became an industry standard
- Motivation – lower pin count & hence lower
 - IC manufacturing/packaging costs
 - PCB complexity & cost
 - Power & size
- Maximal bi-directional data bit rate:
 - Standard-mode: 100 kbps
 - Fast-mode: 400 kbps
 - Fast-mode plus: 1 Mbps
 - High-speed mode: 3.4 Mbps
- Not all devices are capable of all the modes
 - Many low-speed devices support only standard-mode

I2C: COMMON PERIPHERALS

- RTC (Real Time Clock)
- Converters: ADC & DAC
- Memory cards/modules: EEPROM & Flash
- Sensors: accelerometer, temperature, light, pressure sensor etc.
- Signal-mixer, Amplifier, Potentiometer
- LCD/OLED displays
- UART, CAN & USB controller
- SDRAM, DDR SDRAM, DDR2 SDRAM memory stick
- Port expander
- Note: Not every model/make of the peripherals mentioned above use I2C. Some use analog, some use SPI, some use UART etc.
 - For e.g., there are many accelerometers with SPI interface

I2C DEVICES ON THE BOARD

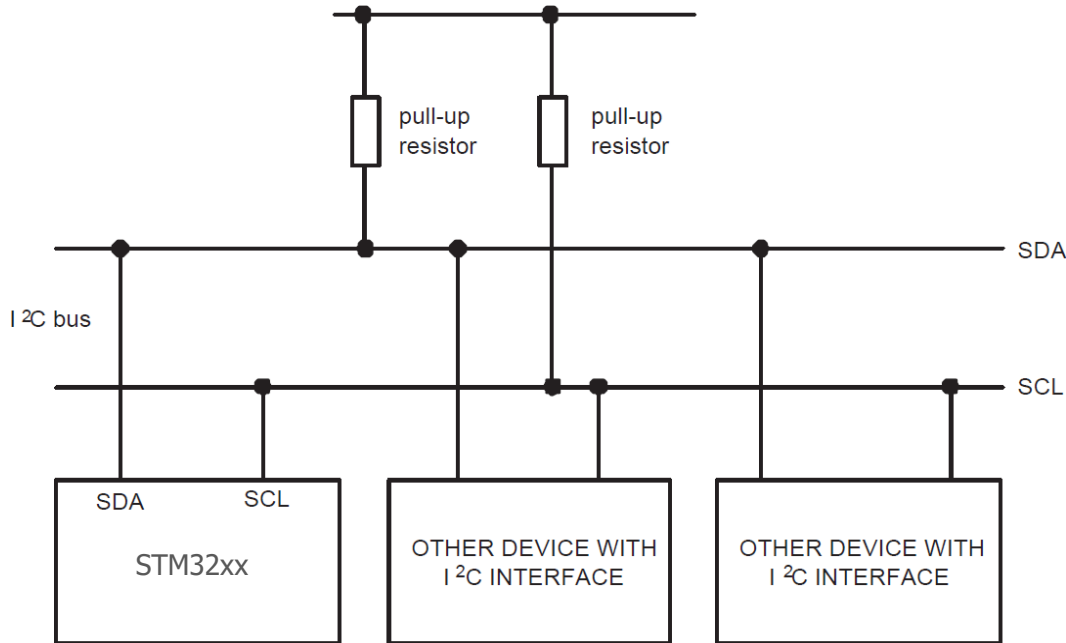


The devices highlighted use I2C2 for data transfer.

They also use GPIOs (EXTIs) for interrupts.

Refer to UM2708: STM32L4+ series Discovery kit, Fig 2 (pg. 7)

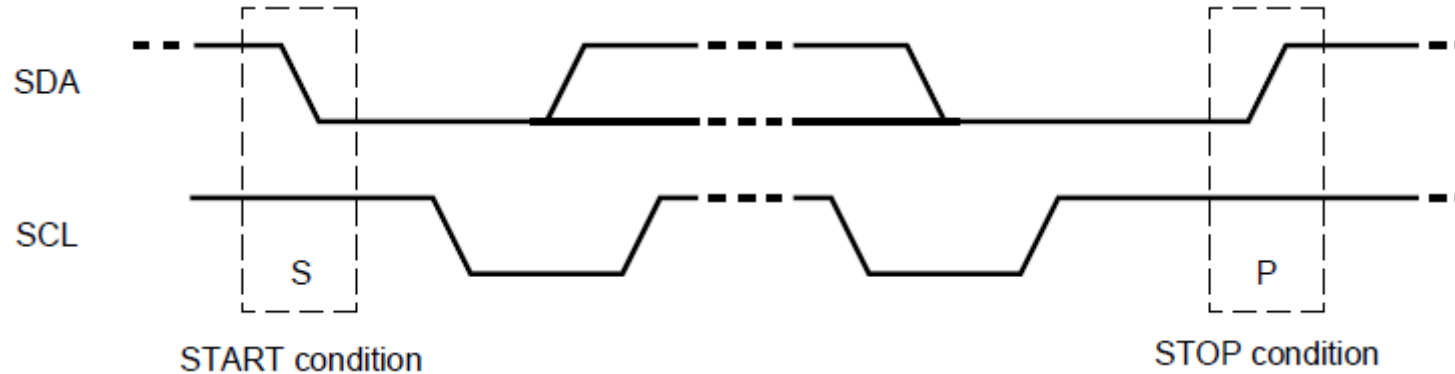
I2C SIGNALS



- Two bi-directional lines:
 - Serial Data line SDA contains the data
 - Serial Clock line SCL with the clock signal for synchronization
- When the bus is free, both lines are high due to pull-up
- The output stages of devices connected to the bus must have an open-drain or open- collector to perform the wired- AND function

Open drain -> a pin outputs **Z** or **low**, never **high**

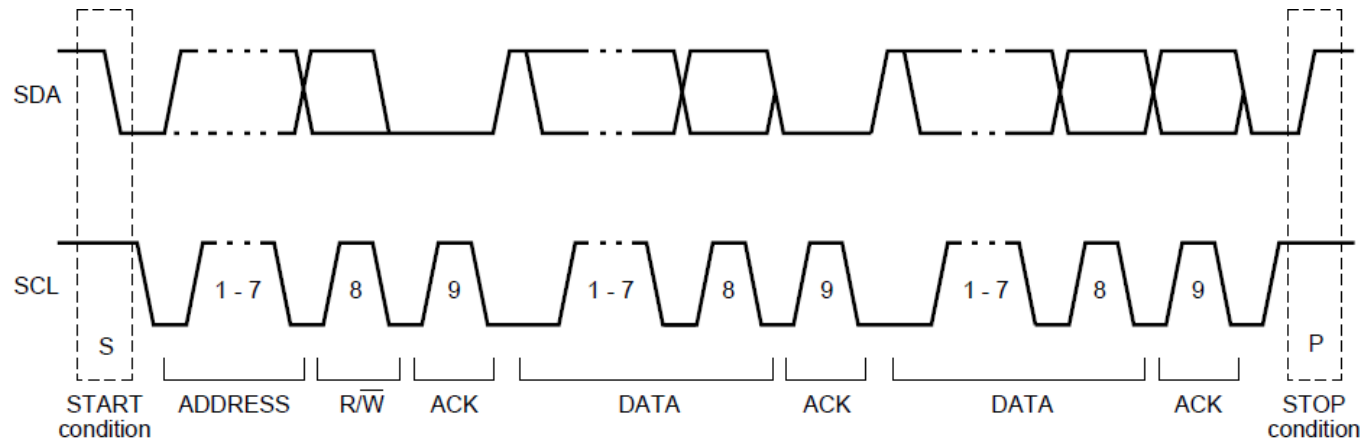
How I2C PROTOCOL WORKS?



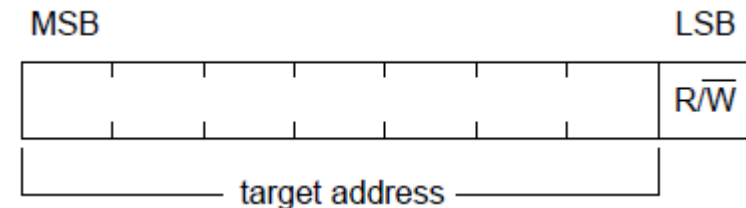
- All transactions begin with a START S and can be terminated by a STOP P, and are always generated by the master
- The bus is considered to be busy after S. The bus is free again a certain time after P
- Another master (if multiple masters are present on the same bus) will not attempt to send a S condition if the bus is not free – all devices will keep listening to the bus all the time to know whether the bus is free or not!
- SDA changes when SCL is high only for S and P. For usual data bits, SDA can change only when SCL is low

Why is S a high->low transition, P a low->high transition, and not the other way around?

How I2C PROTOCOL WORKS?



- After S, a slave address is sent. This address is 7 bits long followed by an eighth bit which is a data direction bit R/W
 - 0 → transmission WRITE
 - 1 → request for data READ
- Sometimes, datasheets mention read (high) and write (low) addresses separately : 7-bit address appended with a 1 (R) / 0 (W)



How I2C PROTOCOL WORKS?

- Every byte put on the SDA line must be 8 bits long
- The number of bytes that can be transmitted per transfer is unrestricted
- The transmitter releases the SDA line during the 9th clock so the receiver can pull the SDA line low, and it remains stable low during the high period of this clock pulse to indicate an ACK (A)
- When SDA remains high during this 9th clock pulse, this is defined as the Not Acknowledge (NACK / $\bar{A}$) signal
- Interpretation depends on whether the master or the slave generates it
- Note : $\bar{A}$ is fact not 'generated'. A lack of A is $\bar{A}$

CONCLUSION

- Characteristics and features of I2C
- How I2C Protocol works?
 - Start & Stop conditions
 - Address byte with 7bit address + R/W bit
 - Acknowledgement and Not Acknowledgement